Methodological Pipeline: Empirical Temporal Link Prediction in Network Security Telemetry

Chronological snapshotting, candidate stratification, negative sampling regimes (Setting A vs. B), and multi-seed evaluation

1. Telemetry Ingestion

Raw Enterprise & Cyber-Range Data

APEX-IDS2026 Telemetry

- Live perimeter ISP infrastructure
- 44 continuous calendar days (2026-06-21 to 2026-08-03)
- 69,107,018 high-confidence flows
- 0 null IPs / 0 NaNs / 53 features
- 412,906 source / 42,421 dest IPs
- CC BY 4.0 Open License
- DOI: 10.5281/zenodo.22013914

CIC-IDS2017 Benchmark

- Cyber-range testbed environment
- 5 consecutive weekdays (Mon–Fri)
- 2,830,743 valid flow records
- 8,832 unique endpoint IPs
- Day 4 (Friday) test transition
- Verified timestamp chronology
- Evaluated alongside APEX to contrast testbed vs. live churn

2. Snapshot Construction

Chronological Graph Partitions

Multi-Scale Windowing

- Primary Scale: Δt = 24h (Daily)
  - APEX: 44 snapshots (T=43 flux)
  - CIC: 5 snapshots (Day 0–4)
- Sub-daily Audit Scales:
  - APEX: Δt ∈ {6h, 12h, 24h}
  - CIC: Δt ∈ {1h, 6h, 24h}
- Partial boundary handling: Day 0 & Day 43 explicitly marked

Directed Graphs G_t = (V_t, E_t)

- Nodes V_t: Active IP endpoints
- Directed Edges E_t: IP flows u → v
- Multi-scale edge novelty: $N_t = |E_{t+1} \setminus E_t| / |E_{t+1}|$
- Symmetric Jaccard turnover: $T_J = |E_{t+1} \Delta E_t| / |E_{t+1} \cup E_t|$

Chronological Splits

- APEX-IDS2026 (44 days):
  - Train: Days 0–24 (56.8%)
  - Validation: Days 25–31 (15.9%)
  - Test: Days 32–43 (27.3%)
- CIC-IDS2017 (5 days):
  - Train: Days 0–2 (Mon–Wed)
  - Validation: Day 3 (Thu)
  - Test: Day 4 (Fri transition)

3. Candidate Sampling

Edge Stratification & Regimes

Target Edge Stratification

Positive Target Edges E_{t+1}:

1. Recurring Edges ($E_t \cap E_{t+1}$):
   - APEX: 46.01% ± 6.82%
   - CIC: 30.94% (9,939 edges)
2. New Between Known ($V_t \times V_t$):
   - APEX: 25.75% ± 2.80%
   - CIC: 38.39% (12,330 edges)
3. Dormant Historical ($V_{<t} \setminus V_t$):
   - APEX: 18.31% ± 3.06%
   - CIC: 21.19% (6,805 edges)
4. Unseen Nodes (V_{new}):
   - APEX: 9.92% | CIC: 9.48%

Negative Sampling Regimes

- Setting A: Hard Historical Negatives
  - Drawn from $E_{\leq t} \setminus E_{t+1}$
  - Inactive historical channels
  - Evaluates link extinction detection
  - Exposes memory baseline collapse
- Setting B: Mixed Random Negatives
  - Non-edges from $V_t \times V_t \setminus E_{t+1}$
  - Unconnected active node pairs
  - Induces 'Sampling Illusion' (≥99% P)
  - Asymmetry limitation: Positives contain ~18-21% dormant nodes

4. Evaluated Predictors

Memory, Heuristics & GNNs

Temporal Memory Baselines

- Persistence Baseline: $\hat{y} = 1 \iff (u, v) \in E_t$
- EdgeBank-∞ (Binary & Frequency): $\hat{y} = 1 \iff (u, v) \in E_{\leq t}$ (Unbounded accumulation)
- EdgeBank-tw (Windowed): $\hat{y} = 1 \iff (u, v) \in \bigcup_{i=0}^{W-1} E_{t-i}$ (Evaluated with W=2 decay)

Topological Heuristics

- Common Neighbors (CN, directed): $s_{CN} = |\Gamma_{out}(u) \cap \Gamma_{in}(v)|$
- Adamic-Adar (AA, undirected): $s_{AA} = \sum_{z \in \Gamma(u) \cap \Gamma(v)} 1 / \log|\Gamma(z)|$
- Jaccard Coefficient (JC): $s_{JC} = |\Gamma(u) \cap \Gamma(v)| / |\Gamma(u) \cup \Gamma(v)|$

Adapted EvolveGCN-H

- Deterministic degree node features (in, out, total, log-degree, ratio)
- Global graph summary: $Z_t = \text{Pool}(X_t)$
- Recurrent weight evolution: $\text{vec}(W_t) = \text{GRU}(Z_t, \text{vec}(W_{t-1}))$
- 2-layer MLP edge link decoder
- Strict torch.no_grad() test state
- Evaluated across 5 random seeds

5. Evaluation Protocols

Threshold & Continuous Metrics

5-Seed Capped Benchmark

- Controlled candidate sets: k = 2,000 per step (N=48,000 APEX) N = 4,000 on CIC Day 4 transition
- Enables tractable evaluation of costly heuristics (AA, JC, CN)
- Balanced positive : negative (1:1)
- Evaluates Setting A vs. Setting B
- 5 fixed seeds: [42,100,123,456,999]
- Mean ± sample std (ddof=1)

Uncapped Test Partition

- N = 3,256,518 candidate pairs (1,628,259 pos + 1,628,259 neg)
- Full evaluation across 12 test steps
- Continuous ROC-AUC & PR-AUC
- EvolveGCN-H: 0.9705 ROC / 0.9716 PR
- Persistence: 0.8130 ROC / 0.8129 PR
- 2×2 empirical confusion matrices

Diagnostic Audits

- EvolveGCN collapse modes (Seeds 123, 456)
- Static whitelist failure: 91.4% false alarms
- Jaccard τ=1.0 cutoff artifact resolved
- Loss curves withheld (no unrecorded logs)